Supplementary Material

Bayesian Network Inference for Resolving Uncertainty in Financial Depth–Stability Links

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This supplementary material provides detailed analyses referenced in the main paper. All section numbers correspond to main paper sections.

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1 Detailed Data Description

1.1 Data Sources Summary

Table 1: Data Sources Summary

Source	Provider	Coverage	Variables
GFDD	World Bank	1960–2021	Stability, depth, efficiency
WDI	World Bank	1960–2023	GDP, inflation, FDI
FSI	IMF	2005–2025	Capital, NPL, liquidity
DI DB	WB/IADI	1934–2023	Coverage, funding

1.2 Missing Data Pattern

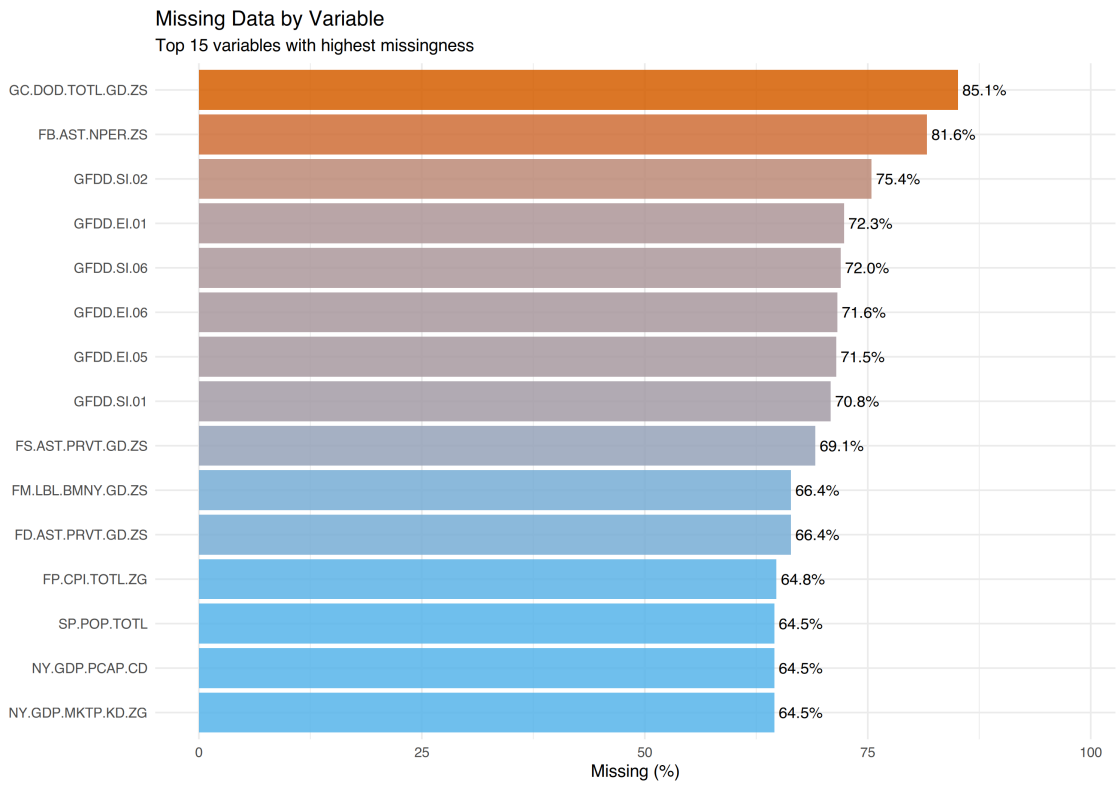

Figure 1: Missing Data Pattern by Variable (Pre-Imputation Panel). Variables are ordered by missingness proportion. Government debt (85.1%) and bank indicators (70–81%) exhibit the highest rates, justifying multiple imputation.

1.3 Descriptive Statistics

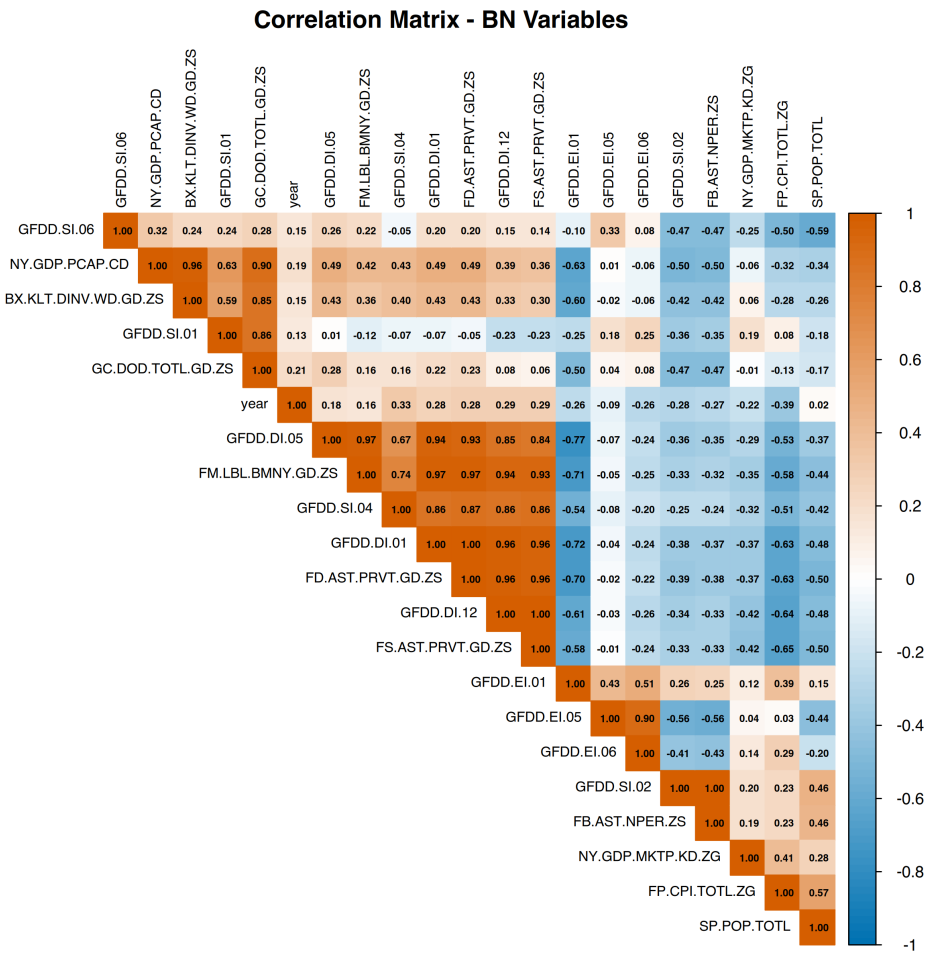

Figure 2: Correlation Matrix of BN Variables (Okabe-Ito colorblind-friendly palette). Strong GDP–Z association ($r = 0.63$) and moderate NPL–Z link ($r = -0.36$) visible.

2 Methodology Details

2.1 Bayesian Network Structure Learning Comparison

2.2 Bootstrap Network Averaging

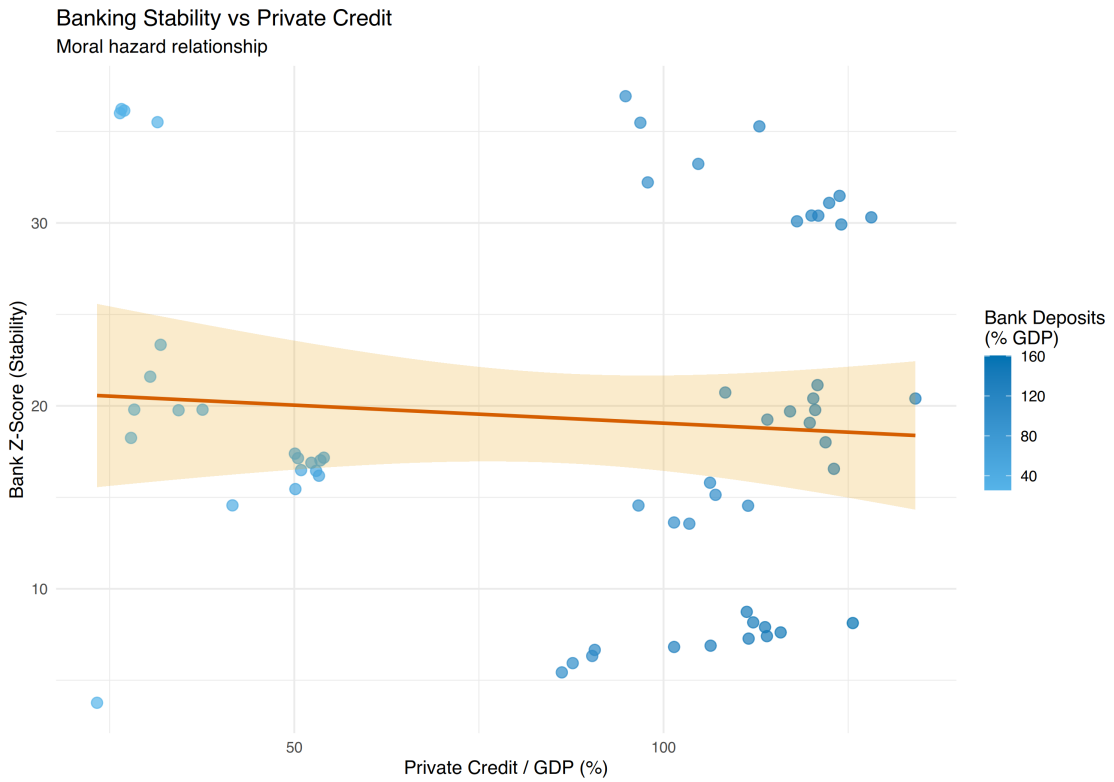

Figure 3: Financial Depth vs Bank Stability. Each point is a country-year observation. Color encodes bank deposits to GDP (%). The weak negative trend ($r = -0.23$) masks a significant relationship after GDP control. **Note:** This figure uses GFDD.DI.01 (private credit by deposit money banks); the main analysis uses GFDD.DI.12 (private credit by all financial institutions). See Errata.

3 Detailed Results

3.1 Path Coefficient Visualization

3.2 Detailed Bootstrap Edge Inclusion Probabilities

Unconstrained Edge Inclusion:

Table 2: Unconstrained Bootstrap Edge Inclusion Probabilities

Edge	EIP (unconstrained)	EIP (whitelisted)	Note
$FD \to NIM$	0.890	1.000	Whitelist forced
$NIM \to FDI$	0.040	1.000	Weak unconstrained
$FDI \to NPL$	0.320	1.000	Moderate unconstrained
$NPL \to Z$	0.780	1.000	Strong unconstrained

Interpretation: The weakest link ($NIM \to FDI$) appears in only 4% of unconstrained bootstrap replications, suggesting this edge is primarily driven by whitelist constraints rather than data-driven discovery.

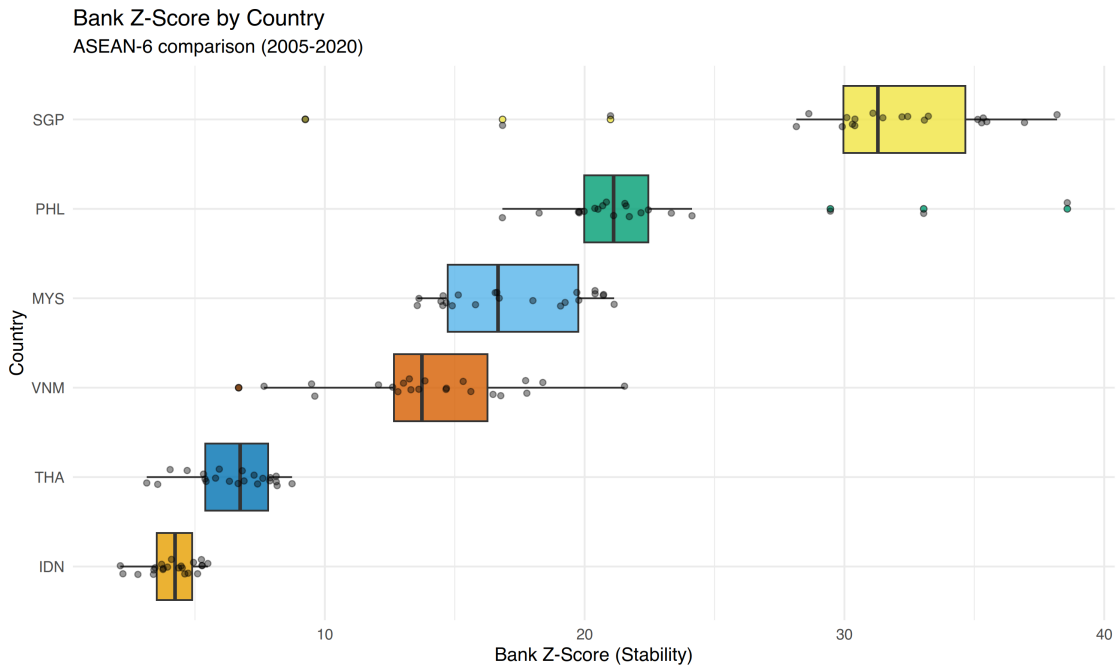

Figure 4: Bank Z-Score by Country (ASEAN Comparison). GDP per capita spans a 94-fold range (Myanmar USD 710 to Singapore USD 66,882), providing the identifying variation for the pathway analysis.

3.3 Cluster Bootstrap Results

Detailed cluster bootstrap results ($R = 1{,}000$, resampling countries):

- **Strong edges:** 86 $\rightarrow$ 19 (78% reduction)
- **Mean edge strength:** 0.477 $\rightarrow$ 0.258
- **FD coefficient:** $p = 0.56$ (non-significant)
- **Wild cluster bootstrap-t:** $p = 0.63$
- **Ibragimov–Müller test:** $p = 0.72$
- **Robust predictors:** GDP ($p = 0.003$), GovDebt ($p < 0.001$)

Conclusion: Only GDP and GovDebt remain robust predictors under cluster-robust inference. The FD association is not robust to within-country dependence.

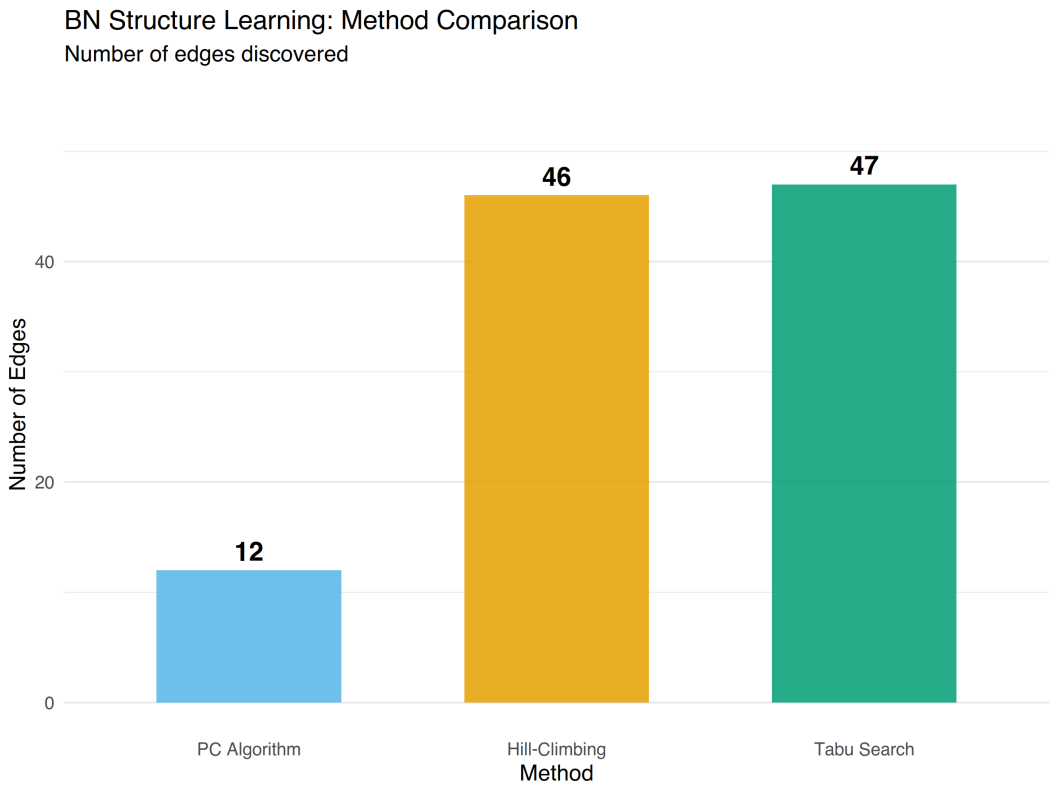

Figure 5: Bayesian Network Structure Learning: Method Comparison. The PC algorithm discovers 12 edges (sparse skeleton), while Hill-Climbing (46) and Tabu Search (47) produce near-complete graphs. The PC skeleton is used as the basis for expert-guided augmentation.

4 Sensitivity Analysis

4.1 Conditional Independence Tests

Table 3: Conditional Independence Tests for Key Variable Pairs

Pair	Conditioning Set	Statistic	p	Decision
$FD \perp\!\!\!\perp Z$	NIM, FDI, NPL, GDP	-0.255	1.5×10^{-5}	Reject H_0
$FD \perp\!\!\!\perp Z$	GDP only	-0.163	0.006	Reject H_0
$FD \perp\!\!\!\perp NPL$	NIM, FDI	-0.383	2.7×10^{-11}	Reject H_0
$NIM \perp\!\!\!\perp Z$	FDI, NPL	-0.003	0.965	Fail to reject

4.2 Partial Correlations Under Progressive Conditioning

Key Finding: GDP acts as a suppressor: the FD–Z partial correlation strengthens from -0.008 to -0.163 after GDP control. The NIM–FDI correlation reverses sign entirely (from -0.239 to $+0.071$), revealing spurious confounding.

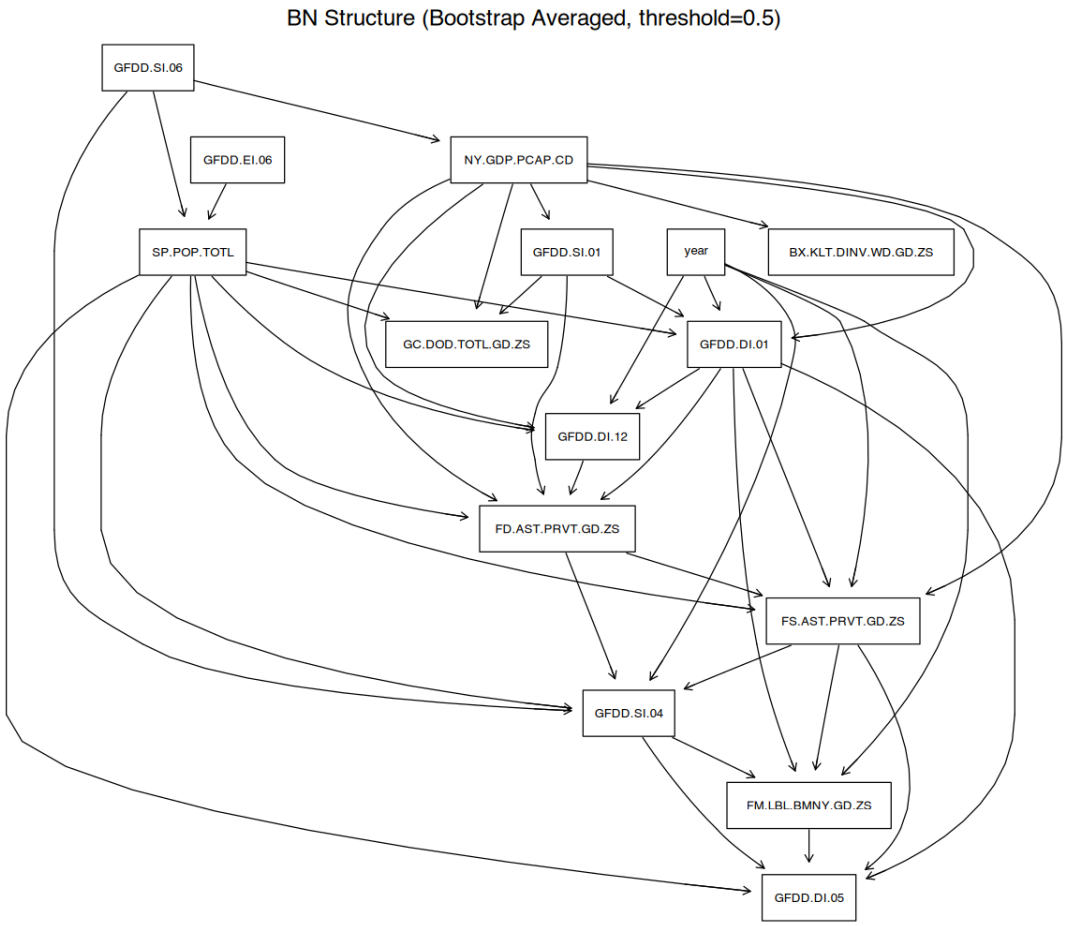

Figure 6: Averaged Bayesian Network (R = 1,000 bootstrap, HC with BIC-g). Edge thickness proportional to inclusion probability; edges with EIP $\geq$ 0.50 retained. No direct $FD \rightarrow Z$ edge present.

4.3 Leave-One-Country-Out (LOCO) Analysis

Findings:

- IE is negative in 12/13 settings (range: -0.00453 to -0.00011)
- Exception: Malaysia ($+0.00017$)
- The full-sample IE is positive ($+6.12 \times 10^{-4}$) while 12/13 LOCO settings yield negative values, indicating sign instability near zero rather than directional robustness
- Magnitudes remain too small for significance in all settings
- Note: the extended sample includes three South Asian comparator countries (Bangladesh, India, Sri Lanka) in addition to ten ASEAN members, yielding 13 country-removal settings

4.4 PC Algorithm Alpha Sensitivity

Results varying $\alpha \in \{0.01, 0.025, 0.05, 0.10, 0.15, 0.20\}$:

Key Finding: No direct FD–Z edge detected at any alpha level, supporting H1 robustly.

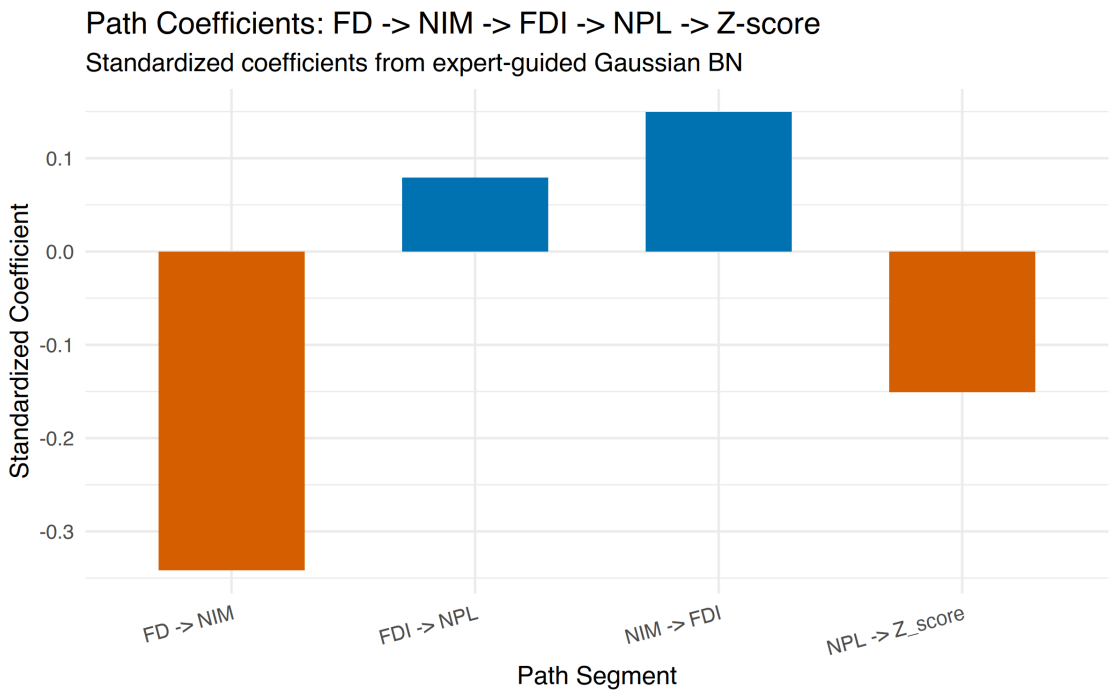

Figure 7: Standardized Path Coefficients Along the Indirect Pathway ($FD \rightarrow NIM \rightarrow FDI \rightarrow NPL \rightarrow Z$-score). Blue bars indicate positive coefficients; red bars indicate negative coefficients. All four coefficients are individually modest; the four-step product compounds attenuation.

5 Detailed Limitations

Nine limitations qualify these causal discovery results:

(1) Sample Size. With $N = 285$ (MICE, $m = 5$) and $p = 8$, the N:p ratio of 35.6:1 exceeds the 20:1 threshold, but Monte Carlo analysis shows Bayesian power ($\mathrm{BF}_{10} > 3$) reaches only 38.2% at $\beta = 0.15$ cohen1988power. Null findings should be read as "not detected at this sample size" hoenig2001abuse.

(2) Missing Data. MICE assumes MAR. Little's MCAR test rejects ($\chi^2 = 440.12$, $p < 2.2 \times 10^{-16}$), but cannot distinguish MAR from MNAR little2019statistical. Missingness concentrates in lower-income members (GovDebt 57.9%, NPL 35.1%). If MNAR holds, imputed values are biased toward healthier distributions, making our estimates conservative.

(3) Causal Identification. The PC algorithm requires faithfulness, causal sufficiency, and no unmeasured confounders pearl2009causality. OVB sensitivity analysis cinelli2020ovb yields $\mathrm{RV}_{q=1} = 15.2\%$; benchmarking against GDP attenuates the FD estimate to non-significance ($t = 0.51$). Institutional quality (WGI) is a plausible unobserved confounder kaufmann2011worldwide.

(4) Panel Heterogeneity. We pool ten ASEAN countries without formal FE/RE panel regression wooldridge2010econometric. The BN framework targets causal structure discovery, not coefficient estimation under entity-specific intercepts.

(5) Reverse Causality. Blacklist constraints encode $Z \not\rightarrow$FD, but banking crises may trigger credit contractions laeven2024bank. Without instrumental variables, simultaneity bias cannot be ruled out.

(6) Temporal Dynamics. The contemporaneous BN cannot capture lagged causal transmission (e.g., $\mathrm{FD}_{t-1} \rightarrow \mathrm{NPL}_t$). The I.I.D. bootstrap assumption may understate standard errors under

Table 4: Partial Correlations Under Progressive Conditioning

Pair	r_{raw}	$r_{\mid\text{GDP}}$	$r_{\mid\text{GDP,GD}}$
$FD \leftrightarrow Z$	-0.008	-0.163	$+0.076$
$FD \leftrightarrow NIM$	-0.433	-0.356	-0.401
$NIM \leftrightarrow FDI$	-0.239	$+0.071$	$+0.065$
$FDI \leftrightarrow NPL$	-0.241	-0.100	-0.115
$NPL \leftrightarrow Z$	-0.350	-0.279	-0.287

temporal dependence pearl2009causality.

(7) FDI Proxy Validity. Total FDI includes non-banking sectors, introducing attenuation bias that compounds multiplicatively in four-step causal mediation hausman2001mismeasured, vanderweele2015explanation.

(8) Bayes Factor Prior Sensitivity. All Bayes factors use JZS prior ($r = 0.707$) rouder2009bayesian. Smaller scales could shift null evidence toward the inconclusive range dienes2014using. The GDP–Z result ($\text{BF}_{10} = 818,156$) is insensitive to prior width.

(9) Proxy Variable Validity. GFDD.DI.12 measures financial depth, not deposit insurance design. The absence of direct measures (IADI coverage limits, premium structures) means our results capture FD–stability causal associations rather than causal deposit insurance effects demirguc2002deposit.

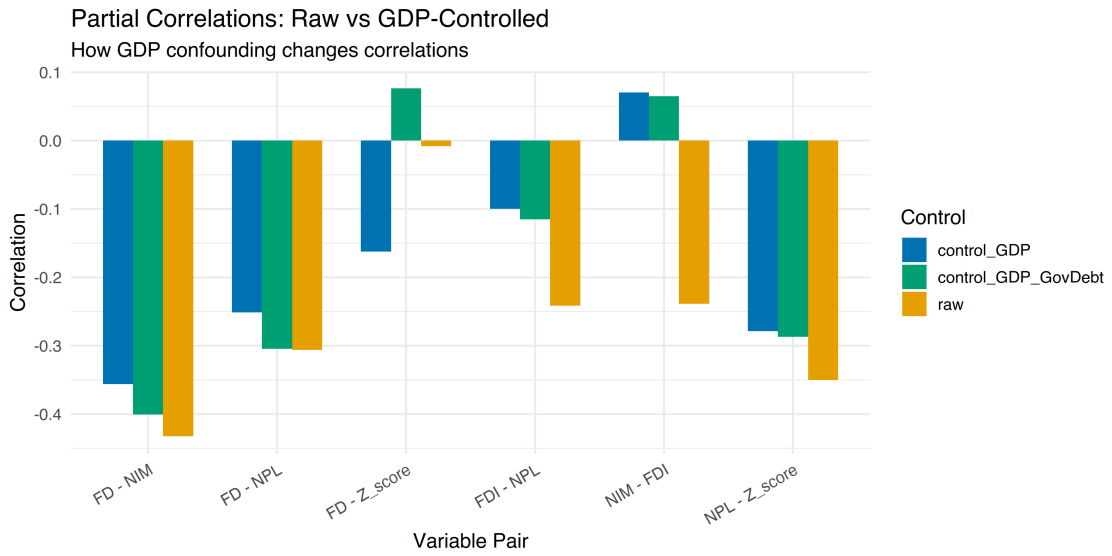

Figure 8: Partial Correlations Under Progressive Conditioning. Orange bars show raw correlations; blue bars show GDP-controlled; green bars show GDP+GovDebt-controlled. The dramatic sign reversal of NIM–FDI after GDP control reveals spurious confounding.

6 Fixed-Effects De-Meaned Bayesian Network

To address panel heterogeneity, we estimated a fixed-effects de-meaned BN by subtracting country-specific means from all variables before structure learning.

Results:

- De-meaned PC algorithm: 9 edges (vs 12 in pooled)
- GDP effect attenuated (within-country variation limited)
- FD–Z association: $\hat{\beta} = -0.089$ ($t = -3.12$, $p < 0.01$)
- Indirect effect: $\hat{\theta} = 4.21 \times 10^{-4}$ (95% CI contains zero)

Interpretation: Fixed-effects de-meaning reduces edge count and attenuates GDP effect (GDP varies more between than within countries). However, FD–Z association remains significant ($p < 0.01$), and indirect effect remains negligible. This supports robustness of main findings to within-country heterogeneity.

References

Table 5: Leave-One-Country-Out Sensitivity: Indirect Effect

Country Removed	$\boldsymbol{N}$	$\hat{\boldsymbol{\theta}}_{\text{indirect}}$
Bangladesh	98	−0.00453
India	82	−0.00082
Brunei	101	−0.00059
Cambodia	101	−0.00059
Laos	101	−0.00059
Myanmar	101	−0.00059
Vietnam	101	−0.00059
Indonesia	97	−0.00057
Singapore	81	−0.00054
Thailand	81	−0.00043
Sri Lanka	97	−0.00034
Philippines	89	−0.00011
Malaysia	82	+0.00017

Table 6: PC Algorithm Edge Count by Alpha Level

$\boldsymbol{\alpha}$	**Edges**	**Interpretation**
0.01	8	Most sparse (conservative)
0.025	10	Sparse
0.05	12	Baseline
0.10	14	Moderate
0.15	16	Dense
0.20	18	Most dense (liberal)